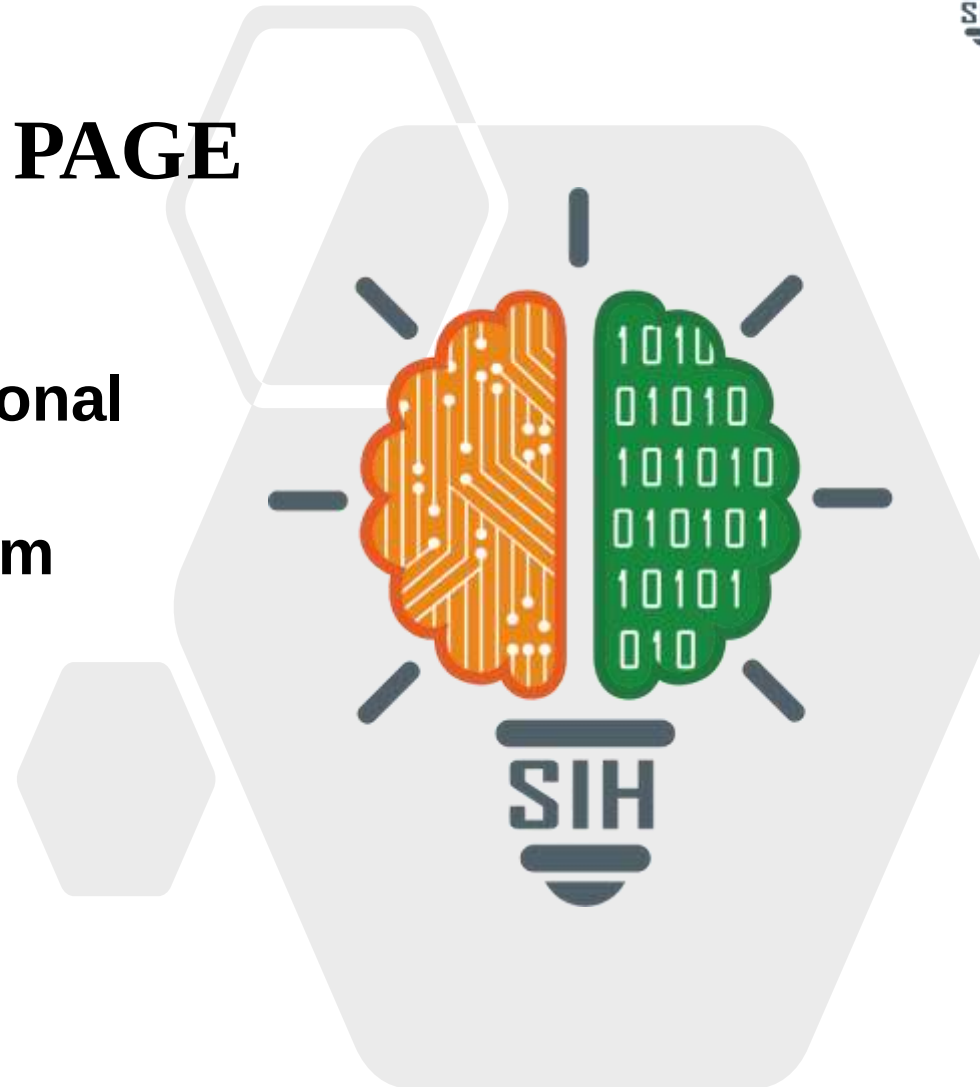


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – 26069
- Problem Statement Title- National Weather Big Data Analytics Platform
- Theme- Disaster Management
- PS Category- Software
- Team ID-
- Team Name - VayuDrishti



National Weather Big Data Analytics

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed Explanation :

- Real time weather data collection from **social media, websites, public datasets and citizen reports.**
- AI/NLP- based processing to extract **location, time, event type & important metadata.**
- ML-based modules for **fake report detection, source verification & duplicate removal.**
- Automatic classification of events: **Floods, Rainfall, Heatwaves, Storms, Fog, Dust Storms, etc.**
- Centralized database with an **interactive real-time dashboard.**

Innovation and Uniqueness :

- 1 Trust score, not a yes/no flag**
Graded credibility an officer can act on
- 2 Cross-modal verification**
Text + EXIF/reverse-image + radar & satellite cross-check
- 3 Hyperlocal & multilingual**
MuRIL handles 12+ Indian languages and Hinglish code-mix
- 4 Crowd as a sensor network**
Fills gaps between sparse AWS stations — often flags an event before the bulletin
- 5 100% open-source stack**
Zero licensing cost; deployable on NIC cloud or MoES on-prem

Technologies Used :

Task	Tools
Frontend & Backend	HTML, CSS, JavaScript, React.js, Python(Flask) or Node.js
Database	MySQL/MongoDB
AI/ML(fake & duplicate detection	Python, Scikit-learn, NLTK(text classification), OpenCV
Hosting/Deployment	GitHub, Render, Heroku(free hosting)
Maps & visualization	Leaflet.js, GoogleMaps API, Charts.js

Methodology :

Point	Covers
Collect & stream data	pulls weather posts, APIs , citizen reports and streams them live
Process & verify with AI	cleans data , tags location, removes duplicates, flags fake reports
Store & display	saves verified data centrally, shows it on live dashboard /admin panel

Feasibility

- Open-source tools → Python, React.js, MongoDB
- Multiple data sources → APIs, websites, datasets & citizen reports
- AI/ML processing → Fake & duplicate report detection Real-time
- dashboard → Live weather event visualization

Key Challenges

- Large volume of real-time
- dataFake, misleading & duplicate
- reports Multilingual/Hinglish
- contentMissing or inaccurate location data

Strategies to Overcome

- Automated data collection & processing pipeline
- ML-based Trust Score for report verification
- NLP & similarity models for duplicate detection
- Cross-verification using multiple weather sources

Disaster Management

Earlier visibility of local events.
Supports faster situational awareness and response.

Citizens & Communities

Simple multilingual reporting.
Useful even where station coverage is sparse.

Government & Agencies

One dashboard for verified reports.
Prioritizes actionable events with trust scores.

Economic & Operational

Open-source stack reduces licensing dependence.
Reusable pipeline across weather-event categories.

Long-term Value

Creates a structured archive of local observations.
Enables analytics for planning and disaster preparedness.

Research basis used for the proposed architecture and data sources:

01

<https://mausam.imd.gov.in/> — India Meteorological Department (IMD): AWS, radar, satellite, warnings & nowcast services

02

<https://www.mosdac.gov.in/> — ISRO / MOSDAC: meteorological & oceanographic satellite data, radar and near-real-time services

03

<https://arxiv.org/abs/2103.10730> — MuRIL: Multilingual Representations for Indian Languages; multilingual / transliterated Indian-language NLP

04

<https://www.sih.gov.in/> — Smart India Hackathon 2026, Problem Statement 26069: National Weather Big Data Analytics Platform

Reference check: IMD and MOSDAC document the availability of AWS / radar / satellite observations and weather services; MuRIL documents Indian-language and transliterated-text handling.